Hyunsung Cho

Human-Computer Interaction Institute
School of Computer Science
Carnegie Mellon University
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Google Scholar

Research Interests

Human-Computer Interaction · Computational User Modeling · Embodied Perception · Extended Reality (XR) · Human-AI Interaction · Adaptive, Generative, and Agentic Interfaces

Education

08/2021 - 05/2027 (expected) **Carnegie Mellon University**
Ph.D. Candidate in Human-Computer Interaction, School of Computer Science
Advisor: David Lindlbauer
Thesis: *Perceptually Intelligent User Interfaces for Spatial Computing* (proposed 05/2026)
Committee: David Lindlbauer, Ken Koedinger, Chris Harrison, Sean Follmer

03/2018 - 02/2020 **Korea Advanced Institute of Science and Technology (KAIST)**
M.S. in Computer Science, School of Computing
Advisor: Sung-Ju Lee
Thesis: *Private Status Sharing and Sender-Controlled Notifications in Mobile Instant Messaging*
Committee: Sung-Ju Lee, Juho Kim, Youn-kyung Lim

08/2013 - 02/2018 **Korea Advanced Institute of Science and Technology (KAIST)**
B.S. in Computer Science (Advanced Major), School of Computing

Awards & Honors

2026 **Link Foundation Modeling, Simulation & Training Fellowship**

2023 **Best Paper Award, ACM ISS**

2021 **Best Paper Award, ACM CSCW**

2021 **Methods Recognition, ACM CSCW**

2023 **10th Heidelberg Laureate Forum Young Researcher** (with SAP Grant)

2020 **NAVER Ph.D. Fellowship**

2026 **XRTC Creative Research Grant**

2024 **Sponsored Research, Meta Reality Labs**

2023 **Sponsored Research, Meta Reality Labs**

2021 **Bell Labs Summer Intern Award for Outstanding Innovation**

2019 **Best Video Award, ACM MobiSys**

2018 **Best Poster/Demo Award, ACM SIGCHI Local Chapter**

2019 **Best TA Award, KAIST School of Computing**

2019 **Google Women Techmakers Scholars**

2016 **Korea SW Hackathon Runner-up, Ministry of Science, ICT and Future Planning**

2016 **Undergraduate Research Program Grant, KAIST**

2014 - 2017 **National Science & Technology Scholarship**

Publications

Conference & Journal Papers

c.22 A Benchmarking Framework for Context-aware XR Interfaces.
Hyunsung Cho, Sarah Yewon Yun, Nancy Ruonan Sun, Ben Lafreniere, Mark Parent, Kashyap Todi, Tanya Jonker, Hrvoje Benko, Tongshuang Wu, and David Lindlbauer.
UIST 2026: ACM Symposium on User Interface Software and Technology

c.21 AtmoSteer: Personalized Scene Control with Reusable Atmospheric Recipes
Nancy Ruonan Sun, **Hyunsung Cho**, and David Lindlbauer.
UIST 2026: ACM Symposium on User Interface Software and Technology

c.20 Simulating Human Audiovisual Search Behavior.
Hyunsung Cho, Xuejing Luo, Byungjoo Lee, David Lindlbauer, and Antti Oulasvirta.
CHI 2026: ACM CHI Conference on Human Factors in Computing Systems

c.19 Augmented Reality Productivity In-the-Wild: A Diary Study of Usage Patterns and Experiences of Working with AR Laptops in Real-World Settings.
Yi Fei Cheng, Ari Carden, **Hyunsung Cho**, Catarina G Fidalgo, Jonathan Wieland, and David Lindlbauer.
TVCG 2025: IEEE Transactions on Visualization and Computer Graphics

c.18 Persistent Assistant: Seamless Everyday AI Interactions via Intent Grounding and Multimodal Feedback.
Hyunsung Cho, Jacqui Fashimpaur, Naveen Sendhilnathan, Jonathan Browder, David Lindlbauer, Tanya R. Jonker, and Kashyap Todi.
CHI 2025: ACM CHI Conference on Human Factors in Computing Systems

c.17 A Dynamic Bayesian Network Based Framework for Multimodal Context-Aware Interactions.
Violet Yinuo Han, Tianyi Wang, **Hyunsung Cho**, Kashyap Todi, Ajoy Savio Fernandes, Andre Levi, Zheng Zhang, Tovi Grossman, Alexandra Ion, Tanya R. Jonker.
IUI 2025: ACM Conference on Intelligent User Interfaces

c.16 Evaluating Dynamic Delivery of Audio+Visual Message Notifications in XR.
Hyunsung Cho, Drew Edgar, David Lindlbauer, Joseph O'Hagan.
IEEE VR 2025: IEEE Conference on Virtual Reality and 3D User Interfaces

c.15 Push2AR: Enhancing Mobile List Interactions Using Augmented Reality.
Jonathan Wieland, **Hyunsung Cho**, Sebastian Hubenschmid, Akihiro Kiuchi, Harald Reiterer, David Lindlbauer.
ISMAR 2024: IEEE International Symposium on Mixed and Augmented Reality

c.14 Auptimize: Optimal Placement of Spatial Audio Cues for Extended Reality.
Hyunsung Cho, Alexander Wang, Divya Kartik, Emily Liying Xie, Yukang Yan, and David Lindlbauer.
UIST 2024: ACM Symposium on User Interface Software and Technology

c.13 SonoHaptics: An Audio-Haptic Cursor for Gaze-Based Object Selection in XR.
Hyunsung Cho, Naveen Sendhilnathan, Michael Nebeling, Tianyi Wang, Purnima Padmanabhan, Jonathan Browder, David Lindlbauer, Tanya R. Jonker, and Kashyap Todi.
UIST 2024: ACM Symposium on User Interface Software and Technology

c.12 MineXR: Mining Personalized Extended Reality Interfaces.
Hyunsung Cho, Yukang Yan, Kashyap Todi, Mark Parent, Missie Smith, Tanya Jonker, Hrvoje Benko, and David Lindlbauer.
CHI 2024: ACM CHI Conference on Human Factors in Computing Systems

c.11 BlendMR: A Computational Method To Create Ambient Mixed Reality Interfaces.
Violet Han, **Hyunsung Cho**, Kiyosu Maeda, Alexandra Ion, and David Lindlbauer.
ISS 2023: ACM Interactive Surfaces and Spaces Conference
🏆 **Best Paper Award**

c.10 FinerMe: Examining App-level and Feature-level Interventions to Regulate Mobile Social Media Use.
Adiba Orzikulova, **Hyunsung Cho**, Hye-Young Chung, Hwajung Hong, Uichin Lee, and Sung-Ju Lee.
CSCW 2023: ACM Conference on Computer Supported Cooperative Work and Social Computing

c.9 RealityReplay: Detecting and Replaying Temporal Changes In Situ using Mixed Reality.
Hyunsung Cho, Matthew L. Komar, and David Lindlbauer.
IMWUT (UbiComp) 2023: ACM Annual Conference on Interactive, Mobile, and Ubiquitous Technologies

c.8 A Survey on Remote Assistance and Training in Mixed Reality Environments.
Catarina Gonçalves Fidalgo, Yukang Yan, **Hyunsung Cho**, Mauricio Sousa, David Lindlbauer, and Joaquim Jorge.
TVCG 2023: IEEE Transactions on Visualization and Computer Graphics

c.7 FLAME: Federated Learning Across Multi-device Environments.
Hyunsung Cho, Akhil Mathur, and Fahim Kawsar.
IMWUT (UbiComp) 2022: ACM Annual Conference on Interactive, Mobile, and Ubiquitous Technologies

c.6 You Are Not Alone: How Trending Stress Topics Brought #Awareness and #Resonance on Campus.
Ryuhaerang Choi, Chanwoo Yun, **Hyunsung Cho**, Hwajung Hong, Uichin Lee, and Sung-Ju Lee.
CSCW 2022: ACM Conference on Computer Supported Cooperative Work and Social Computing

c.5 Prediction for Retrospection: Integrating Algorithmic Stress Prediction into Personal Informatics Systems for College Students' Mental Health.
Taewan Kim, Haesoo Kim, Ha Yeon Lee, Hwarang Goh, Shakhboz Abdigapporov, Mingon Jeong, **Hyunsung Cho**, Kyungsik Han, Youngtae Noh, Sung-Ju Lee, and Hwajung Hong.
CHI 2022: ACM CHI Conference on Human Factors in Computing Systems

c.4 Reflect, not Regret: Understanding Regretful Smartphone Use with App Feature-Level Analysis.
Hyunsung Cho, DaEun Choi, Donghwi Kim, Wan Ju Kang, Eun Kyoung Choe, and Sung-Ju Lee.
CSCW 2021: ACM Conference on Computer Supported Cooperative Work and Social Computing
🏆 **Best Paper Award &** 🏆 **Methods Recognition**

c.3 I Share, You Care: Private Status Sharing and Sender-Controlled Notifications in Mobile Instant Messaging.
Hyunsung Cho, Jinyoung Oh, Juho Kim, and Sung-Ju Lee.
CSCW 2020: ACM Conference on Computer Supported Cooperative Work and Social Computing

c.2 Knocker: Vibroacoustic-based Object Recognition with Smartphones.
Taesik Gong, **Hyunsung Cho**, Bowon Lee, and Sung-Ju Lee.
IMWUT (UbiComp) 2019: ACM Annual Conference on Interactive, Mobile, and Ubiquitous Technologies

c.1 Intelligent Positive Computing with Mobile, Wearable, and IoT Devices: Literature Review and Research Directions.
Uichin Lee, Kyungsik Han, **Hyunsung Cho**, Kyong-Mee Chung, Hwajung Hong, Sung-Ju Lee, Youngtae Noh, Sooyoung Park, and John M. Carroll.
Ad Hoc Networks Journal, Volume 83, 2019

Doctoral Symposium

p.9 Perceptually Intelligent UI for Augmented Everyday Interaction.
Hyunsung Cho
UIST 2025 Doctoral Symposium

Posters, Demos, and Workshop Papers

p.8 MineXR: Mining Personalized Extended Reality Interfaces.
Hyunsung Cho, Yukang Yan, Kashyap Todi, Mark Parent, Missie Smith, Tanya Jonker, Hrvoje Benko, and David Lindlbauer.
CHI 2024 Workshop: Computational Methodologies for Understanding, Automating, and Evaluating User Interfaces

p.7 Evaluating Adaptive XR Systems.
Hyunsung Cho, Yi Fei Cheng, Yukang Yan, and David Lindlbauer.
CHI 2023 Workshop: The Future of Computational Approaches for Understanding and Adapting User Interfaces

p.6 Facilitating Instant Interactions for Stressful Experiences Sharing and Peer Support.
Ryuhaeraeng Choi, Chanwoo Yun, **Hyunsung Cho**, Hwajung Hong, Uichin Lee, and Sung-Ju Lee.
MobiSys 2022 Demos: ACM International Conference on Mobile Systems, Applications and Services

p.5 Device or User: Rethinking Federated Learning in Personal-Scale Multi-Device Environments.
Hyunsung Cho, Akhil Mathur, and Fahim Kawsar.
AIChallengeIoT 2021: ACM SenSys 2021 Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things

p.4 I Share, You Care: Private Status Sharing and Sender-Controlled Notifications in Mobile Instant Messaging.
Hyunsung Cho, Jinyoung Oh, Juho Kim, and Sung-Ju Lee.
CSCW 2020 Demos: ACM Conference on Computer Supported Cooperative Work and Social Computing

p.3 Sender-Controlled Mobile Instant Message Notifications Using Activity Information.
Hyunsung Cho, Jinyoung Oh, Juho Kim, and Sung-Ju Lee.
MobiSys 2019 Demos: ACM International Conference on Mobile Systems, Applications and Services

p.2 Real-Time Object Identification with a Smartphone Knock.
Taesik Gong, **Hyunsung Cho**, Bowon Lee, and Sung-Ju Lee.
MobiSys 2019 Videos: ACM International Conference on Mobile Systems, Applications and Services
🏆 **Best Video Award**

p.1 Identifying Everyday Objects with a Smartphone Knock.
Taesik Gong, **Hyunsung Cho**, Bowon Lee, and Sung-Ju Lee.
CHI 2018 Extended Abstract: ACM Conference on Human Factors in Computing Systems

Patents

2025 Method and Device for Analyzing Feature-Level Usage of App. Korean Patent (Granted: 2025.04.29, No.10-2803674)
Sung-Ju Lee, **Hyunsung Cho**, Donghwi Kim, Daeun Choi.

2025 Audio-Haptic Cursor for Assisting with Virtual or Real-World Object Selection in Extended-Reality (XR) Environments, and Systems and Methods of Use Thereof. U.S. Patent App. (Publication Date: 2025.03.27, No.18/896727)
Kashyap Todi, Tanya Renee Jonker, **Hyunsung Cho**, Naveen Sendhilnathan, Tianyi Wang.

2019 Method and Apparatus for Identifying Object Based on Knock. Korean Patent (Granted: 2019.11.27, No.10-2051493)
Sung-Ju Lee, Taesik Gong, **Hyunsung Cho**, Bowon Lee.

Professional Experience

08/2021 - **Carnegie Mellon University,** Pittsburgh, PA, USA
Research Assistant with David Lindlbauer

05/2026 - **Google,** San Jose, CA
Student Researcher with Xun Qian
Agentic XR experiences and benchmarks for XR coding agents

06/2025 - 09/2025 **Aalto University,** Espoo, Finland
Visiting Researcher with Antti Oulasvirta
Computational model of human audiovisual search behavior [c.20]

05/2024 - 09/2024, 05/2023 - 09/2023
Meta Reality Labs, Redmond, WA, USA
Research Scientist Intern with Kashyap Todi
2024: AI assistant interaction via intent grounding and multimodal feedback [c.18]
2023: Audio-haptic feedback for gaze-based object selection [c.13]

05/2021 - 07/2021
Nokia Bell Labs, Cambridge, UK
Research Scientist Intern with Akhil Mathur
Federated learning across multi-device environments [c.7, p.5]

03/2017 - 04/2021
KAIST Networking and Mobile Systems Lab (NMSL), Daejeon, South Korea
Research Assistant with Sung-Ju Lee

06/2017 - 08/2017
Kiswe Mobile Inc., New Providence, NJ, USA
Web Frontend Developer
Developed an interactive trivia widget featured in VRT Sporza's live IAAF Diamond League stream. Also built streamer tools including a multi-cam layout editor, drag-and-drop thumbnail uploader, and live stream control interface.

12/2015 - 02/2017
KAIST Interaction Lab (KIXLAB), Daejeon, South Korea
Research Assistant with Juho Kim & Jihee Kim

Teaching

Guest Lecturer & Student Panel

Spring 2026
17-722 Human-AI Interaction Using Sensing Systems, Carnegie Mellon University
Instructors: Mayank Goel & Yuvraj Agarwal

Fall 2025
05-499 Interactive Extended Reality, Carnegie Mellon University
Instructor: David Lindlbauer

Spring 2025
05-499B Computational Methods for Interactive Systems, Carnegie Mellon University
Instructor: Alexandra Ion

Fall 2024
CSC 211 Introduction to Human-Computer Interaction, University of Rochester
Instructor: Yukang Yan

Fall 2024
SI 559 Introduction to AR/VR Application Design, University of Michigan
Instructor: Michael Nebeling

Spring 2019
CS492 Introduction to Research, KAIST
Instructors: Juho Kim, Sung-Ju Lee, and Shin Yoo

Teaching Assistant

Fall 2022
05-430 Programming Usable Interfaces, Carnegie Mellon University
Instructor: Alexandra Ion

Spring 2022
05-391 Designing Human-Centered Software, Carnegie Mellon University
Instructor: Chris Harrison

Spring 2019
Head TA, CS341 Introduction to Computer Networks, KAIST
Instructor: Sung-Ju Lee

Spring 2020, Fall 2018
CS341 Introduction to Computer Networks, KAIST
Instructor: Sung-Ju Lee

Fall 2020, Fall 2017, Spring 2017, Fall 2016
CS101 Introduction to Programming, KAIST

Invited Talks

02/2026	ColumbiaHCI Seminar
02/2026	NYU Immersive Computing Lab (Host: Qi Sun)
11/2025	Stanford HCI Lunch
11/2025	Stanford SHAPE Lab (Host: Sean Follmer)
11/2025	Google (Hosts: Ruofei Du & Minsuk Chang)
05/2025	Miraikan Accessibility Lab (Host: Xiyue Wang)
02/2025	Computational Behavior Lab, Aalto University (Host: Antti Oulasvirta)
01/2025	Human-centered Computer Systems Lab, Seoul National University (Host: Youngki Lee)
12/2024	KAIST (Host: Juho Kim)
10/2024	VASC Seminar, Carnegie Mellon University
01/2024	Networking & Mobile Systems Lab, KAIST (Host: Sung-Ju Lee)
11/2019	Information Accessibility Technology (IAT) Conference, Seoul, South Korea

Service

Program Committee

Program Committee Member:	ACM CHI 2027 ACM SIGGRAPH Asia 2026 Technical Communications and Posters ACM CHI Late Breaking Work 2025, 2024
Program Committee Chairing:	ACM UIST 2027, 2026 Posters Co-chair ACM CHI 2024 Subcommittee Chair Assistant - “Interacting with Devices”

Conference Organizing

Organizing Committee:	ACM UIST 2027, 2026 Posters Co-chair ACM UIST 2025 Registration Co-chair
Session Chair:	ACM UIST 2025: Session “Interacting with Audio and Video”
Student Volunteer:	ACM CHI 2024 ACM UIST 2021, 2022 ACM MobiSys 2019

Reviewing

Conferences:	ACM CHI 2022, 2023, 2024, 2025, 2026 ACM UIST 2022, 2023, 2024, 2025, 2026 IEEE ISMAR 2024, 2025, 2026 ACM DIS 2025, 2026 ACM VRST 2026 ACM NordiCHI 2026 IEEE VR 2025 EuroXR 2025 ACM MobileHCI 2021, 2024 ACM SUI 2024
Journals:	ACM Transactions on Computer-Human Interaction (TOCHI) 2026 IEEE Transactions on Visualization and Computer Graphics (TVCG) 2026 ACM IMWUT 2022, 2023, 2024, 2025, 2026 Proceedings of the ACM on Human-Computer Interaction/CSCW 2020, 2021, 2026 ACM Transactions on Applied Perception 2025 International Journal of Human–Computer Interaction 2022 PRESENCE: Virtual and Augmented Reality 2022 ACM Computing Surveys 2020

Posters:	ACM SIGGRAPH Poster 2022, 2023, 2025 ACM CHI Late breaking Works 2021 ACM MobileHCI Poster 2021

Recognized for Excellence in Reviewing

2020 - 2026:	3×UIST 2026, 2×CHI 2026, NordiCHI 2026, 2×UIST 2025, 3×CHI 2025, 2×IMWUT 2025, 3×UIST 2024, CHI LBW 2024, IMWUT 2023, CHI 2023, CHI 2022, CSCW 2021, CSCW 2020

University/Department Service

2022, 2024, 2025	CMU SCS Graduate Application Support Program (GASP)
2023 - 2024	CMU SCS Women/Non-binary Lunch Organizer

Student Advising

2026	Ryan Zhang, BS student at CMU
2025 - 2026	Ollie Arrison, BS student at CMU; now a MS student at CMU
2024 - 2026	Nancy Ruonan Sun, BS student at CMU; now a MS student at CMU
2024 - 2026	Sarah Yewon Yun, BS→MS student at CMU
2024	Anika Bhagavatula, BS student at CMU; now a MS student at UC Berkeley
2022 - 2024	Divya Kartik, BS student at CMU; now at Jump Trading
2023	Alexander Wang, MS student at CMU; now a Ph.D. student at CMU
2023	Emily Xie, BS student at CMU; now at Apple
2022	Violet Han, MS student at CMU; now a Ph.D. student at CMU
2022	Kiyosu Maeda, MS student at UTokyo; now a Ph.D. student at Princeton
2021	Matthew Komar, BS student at CMU; now at Applied Intuition
2021	Adiba Orzikulova, MS student at KAIST; now a Ph.D. candidate at KAIST
2021	Ryuhaerang Choi, MS student at KAIST; now a Postdoc at KAIST
2020	Daeun Choi, BS student at KAIST; now a Ph.D. student at KAIST
2019	Jinyoung Oh, BS student at KAIST; now at Furiosa AI
2018	Chan Ju (Martin) Chong, BS student at KAIST; now MS student at UIUC

Exhibitions & Selected Media Coverage

Exhibitions

03/2022	AI in Weird Wonderland Celine Park Gallery, Seoul, South Korea
2018 -	Knocker: Vibroacoustic-based Object Recognition with Smartphones KAIST Vision Hall, Daejeon, South Korea

Selected Media Coverage

04/2026	Carnegie Mellon University, Extended Reality Technology Center. *Simulating Human Audiovisual Search Behavior*
09/2022	KAIST Breakthroughs. *Can AI empower college students to be their own health agent?*
11/2021	Carnegie Mellon University, School of Computer Science. *CMU Researcher Seeks To Understand the Regret Behind Social Media*
11/2021	Tech Xplore. *Researcher seeks to understand the regret behind social media use*

11/2021 EurekAlert. *Carnegie Mellon University researcher seeks to understand the regret behind social media*

10/2019 Electronics Weekly. *Sensor fusion lets phone identify objects by simply knocking against them*

10/2019 ScienceDaily. *Object identification and interaction with a smartphone knock*

10/2019 New Atlas. *Smartphone tech recognizes objects by being knocked against them*

04/2018 Nerdiest. *This algorithm makes smartphones recognize objects just by Knocking them*

04/2018 ICT NEWS. *New application allows you to identify an object by tapping it with a smartphone*

08/2017 Sporza. *Sporza biedt interactieve primeur met Kiswe Mobile tijdens Diamond League Brussel*

08/2017 PRWeb. *VRT Sporza and Kiswe Offer End Users a First Ever Fully Interactive Mobile Experience Around Diamond League Brussels*

08/2017 VRT Innovatie. *Interactieve Primeur Tijdens Diamond League Brussel Met VRT Sandbox, Sporza & Kiswe*

References

David Lindlbauer
Assistant Professor, School of Computer Science, Carnegie Mellon University
davidlindlbauer@cmu.edu

Hrvoje Benko
Affiliate Professor, Information School, University of Washington; Former Director of Research, Meta Reality Labs
benko@uw.edu

Antti Oulasvirta
Professor, School of Electrical Engineering, Aalto University, Finland
antti.oulasvirta@aalto.fi

Sean Follmer
Associate Professor, School of Engineering, Stanford University
sfollmer@stanford.edu